

Final Project Report

1. Introduction

1.1. Project overviews:

This project focuses on understanding the patterns and trends in India's agricultural cultivation from 1997 to 2021.

The project uses historical agricultural data to study crop production across different regions and seasons.

Tableau is used to convert complex agricultural data into simple and interactive visualizations.

The dashboards show important information about crop production, seasonal changes, regional performance, and overall trends.

These visualizations make it easier to compare crops and identify high- and low-performing regions.

The project can help farmers make better decisions about crop selection and cultivation.

It can also support government authorities and investors in understanding the agricultural sector.

Overall, the project aims to turn agricultural data into useful insights for better decision-making.

1.2. Objectives:

The main objective of this project is to analyze India's agricultural cultivation data from 1997 to 2021.

It aims to identify the crops that have performed well over different years and seasons. Another objective is to compare agricultural productivity across different states and regions of India.

The project also studies seasonal variations to understand how different seasons affect crop production.

It aims to identify long-term growth and declining trends in agricultural production.

The visualizations help farmers select suitable crops and reduce the risk of poor production.

The project also provides useful insights for government policies, resource allocation, and agricultural investment.

Overall, the objective is to use Tableau-based data visualization to support simple, effective, and data-driven agricultural decisions.

2. Project Initialization and Planning Phase:

2.1. Define Problem Statement:

Project Initialization and Planning Phase

Date	14 august 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Scenario 1: Small-Scale

Farmer

Rajesh is a progressive small-scale farmer based in Uttar Pradesh aiming to maximize crop yield and farm revenue. Unpredictable monsoon shifts and volatile market demands make seasonal crop planning challenging. Through historical data insights (1997–2021), Rajesh seeks seasonal trends, soil-appropriate high-performing varieties, and regional productivity benchmarks to make informed, risk-mitigated farming decisions.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Beca use	Which makes me feel
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PS-1	Small-Scale Farmer	I want to maximize my crop yield and income.	I Struggle to decide which crops to grow each season due to changing weather patterns and market demand	I want to maximize my yield	I must analyze historical agricultural data (1997–2021), I need insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk.
PS-2	Government Policy Analyst	I am designing policies to improve agricultural productivity across India	I need to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations	I am working with the Ministry of Agriculture	By using Tableau dashboards, I am trying to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively.

2.2. Project Proposal (Proposed Solution):

Project Initialization and Planning Phase

Date	16 august2026
Team ID	
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Project Proposal (Proposed Solution)

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Visualize Trends: Identify multi decades crop output cycle, seasonal variations , and state level productivity. • Empower Stakeholders: Provide prescriptive guidance for farmers, policymakers, and agribusiness investors. Build Interactive Dashboards: Construct intuitive Tableau stories, map overlays, and drill down interfaces. Optimize Yields: Highlight top-performing crops and isolate low-yield zones for focus ed intervention.
Scope	• Data Preparation: Ingestion, cleaning, and standardization of 1997–2021 agricultural data from open repositories. Geographical & Seasonal Coverage: Spans 30+ states, Union Territories, and all cropping seasons (Kharif,

	Rabi, Summer, Autumn, Whole Year Visual Engineering: Development of dynamic Tableau worksheets, KPI metrics, geospatial maps, and user filters Performance & Documentation: Optimized data extraction.
Problem Statement	
Description	Analyzes two decades of Indian agricultural data (1997–2021) to map regional productivity and crop performance. Cleans and transforms complex multi-year agricultural datasets into standardized analytical formats. Leverages interactive Tableau dashboards to visually track seasonal yield fluctuations and regional disparities. Evaluates trends across diverse cropping seasons (Kharif, Rabi, Whole Year) and major Indian states.
Impact	For Farmers: Reduces financial and climatic vulnerability via validated crop-yield insights. • For Policymakers: Facilitates targeted agricultural credit, irrigation planning,. • For Agribusinesses: Decreases market uncertainty through reliable crop supply forecast
Proposed Solution	
Approach	Data Ingestion & Cleaning: Aggregate multi-decade raw datasets (1997–2021), clean duplicate/null records, and normalize key variables such as production quantity, cultivated area, and yield metrics. Exploratory Analytics: Perform statistical analysis to categorize regional performance, identify dominant crop varieties, and isolate long-term trends across different agricultural zones. Visual Data Architecture: Build custom dynamic calculations and hierarchical data structures within Tableau to support cross-filtering between states, districts, seasons, and years. Dashboard Design & Deployment: Construct

	specialized interactive stories and drill-down visual interfaces tailored to farmers, policy analysts, and agribusiness investors.
Key Features	Geospatial Yield Mapping: Interactive state and district maps displaying crop density, yield distribution, and regional production highlights. Multi-Year Trend Analysis: Line and area charts tracking production volume and land usage over 24+ years to map growth trajectories. Season-Wise Drill Downs: Comparative visual matrices isolating Kharif, Rabi, Summer, and Whole Year cropping patterns. Interactive Filtering: Cross-functional KPI cards, drop-down filters, and top N state/crop visual summaries for dynamic stakeholder exploration.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git

Data		
Data	Source, size, format	Kaggle dataset, 10,000 images

2.3. Initial Project Planning:

Initial Project Planning Template

Date	15 august2026
Team ID	
Project Name	India's Agricultural Crop Production
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Function al Require ment (Epic)	User Story Numb er	User Story / Task	Story Point s	Priori ty	Team Membe rs	Sprin t Start Date	Sprint End Date (Planned)
Sprint-1	Data collectio n	USN-1	Downloading Dataset	2	High	Balwant Singh	15 Augu st	20 August
Sprint-1	Data collectio n	USN-2	Loading the dataset	1	Low	Balwant Singh	15Au gust	20 August
Sprint-1	Data preparat ion	USN-3	Clean the data	2	High	Balwant Singh	15 Augu st	20 August
Sprint-1	Data Visualiz ation's	USN-4	Developing visualizations	2	Medi um	Balwant Singh	15 Augu st	20 August
Sprint	Function	User	User Story / Task	Story	Priori	Team	Sprin	Sprint End

	al Require ment (Epic)	Story Numb er		Point s	ty	Membe rs	t Start Date	Date (Planned)
Sprint-2	Dashbo ard	USN-5	. Creating Dashboard	2	High	Balwant Singh	22 Augu st	28 August
Sprint-2	story	USN-6	Creating story	1	High	Balwant Singh	22 Augu st	28 August
Sprint-2	Docume ntation and recordin g	USN-7	Creating Demo	2	Low	Balwant Singh	22 Augu st	28 August
Sprint-2	Docume ntation and recordin g	USN-8	Creating Document	2	Medi um	Balwant Singh	22 Augu st	28 August

3. Data Collection and Preprocessing Phase:

3.1. Data Collection Plan and Raw Data Sources Identified:

Data Collection and Preprocessing Phase

Date	13 August2026
Team ID	-----
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification : -

Data collection:

Section	Description
Project Overview	The project develops an interactive business intelligence architecture to analyze multi-decade agricultural production across 30+ states and 600+ districts (1997–2021). The resulting Tableau dashboards provide clear insights to assist stakeholders in boosting productivity and minimizing risk.
Data Collection Plan	• Extract multi-year district and state agricultural records from government open portals. • Capture spatial attributes (State, District), temporal attributes (Crop Year, Season), and production figures (Area in Hectares, Production Quantity). • Ingest and validate data schemas into structured analytical tables for high-performance querying.
Raw Data Sources Identified	Raw Data Sources Identified Ministry of Agriculture (DES / APY Corpus): Official Area, Production, and Yield statistics across 30+ states, 56+ crops, and 5 cropping seasons (1997–2021).

Raw Data Source:

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Ministry of agriculture and farmers welfare.	https://agriwelfare.gov.in/	CSV format dataset	1GB	Public
Dataset 2	Smart bridge site	thesmartbridge.com	Excel	1 GB	Public
Dataset 3	Official statics compiled by the directorate of economics and statics	https://desagri.gov.in/	Excel	1GB	Public

3.2. Data Quality Report: Data Collection and Preprocessing Phase

Date	14 August 2026
Team ID	
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Data Quality Report:

Data Source	Data Quality Issue	Severity	Resolution Plan
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Dataset 01	Numerical entries are absent.	Moderate	Filter dataset and impute absent numerical values.
Dataset02	Crop- separated production report wise	High	Make a file containing all reports

3.3. Data Exploration and Preprocessing:

Data Collection and Preprocessing Phase

Date	17August 2026
Team ID	---
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing: -

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	Schema Inspection: Examined dataset attributes including, Season, Crop, Area (Hectares), and Production. Dataset Profiling: Assessed missing values, structural irregularities, and value distributions across 30+ states, 600+ districts, and 50+ crop varieties from 1997 to 2021.

Data Cleaning	Data Cleaning Handled null values in production through localized median yield estimation; dropped corrupted records with invalid coordinates or zero-area entries. Removed duplicate rows and noise caused by legacy manual data entry.
Data Transformation	Calculated the standardized productivity metric . Normalization: Grouped low-frequency minor crops into broad categories (e.g., Pulses, Oilseeds, Cereals) for macro-level trend analysis.
Data Type Conversion	Data Type Conversion Cast Area and Production fields to 64-bit floating point for aggregation accuracy; converted Crop Year to discrete temporal values and encoded categorical dimensions into structured string objects.
Column Splitting and Merging	Standardization: Cleaned and merged inconsistent state/ district naming variations resulting from geographical boundary redefinitions over time. Attribute Parsing: Split combined seasonal text attributes (e.g., combining climate tags) into standardized season labels (Kharif, Rabi, Summer, Whole Year).
Data Modeling	Data Modeling Designed an optimized Star Schema with a centralized fact table (Crop Production Fact) linked to lookup dimensions (Time Dimension, Geography Dimension, Crop Dimension). rapid Tableau query execution.
Save Processed Data	Storage Format: Exported clean data to optimized .csv / . parquet formats and pushed structured tables to SQL database staging tables. Tableau Extraction: Created a hyper-extracted data source file (. hyper) ready for rapid visualization render in

4. Data Visualization:

4.1. Framing Business Questions:

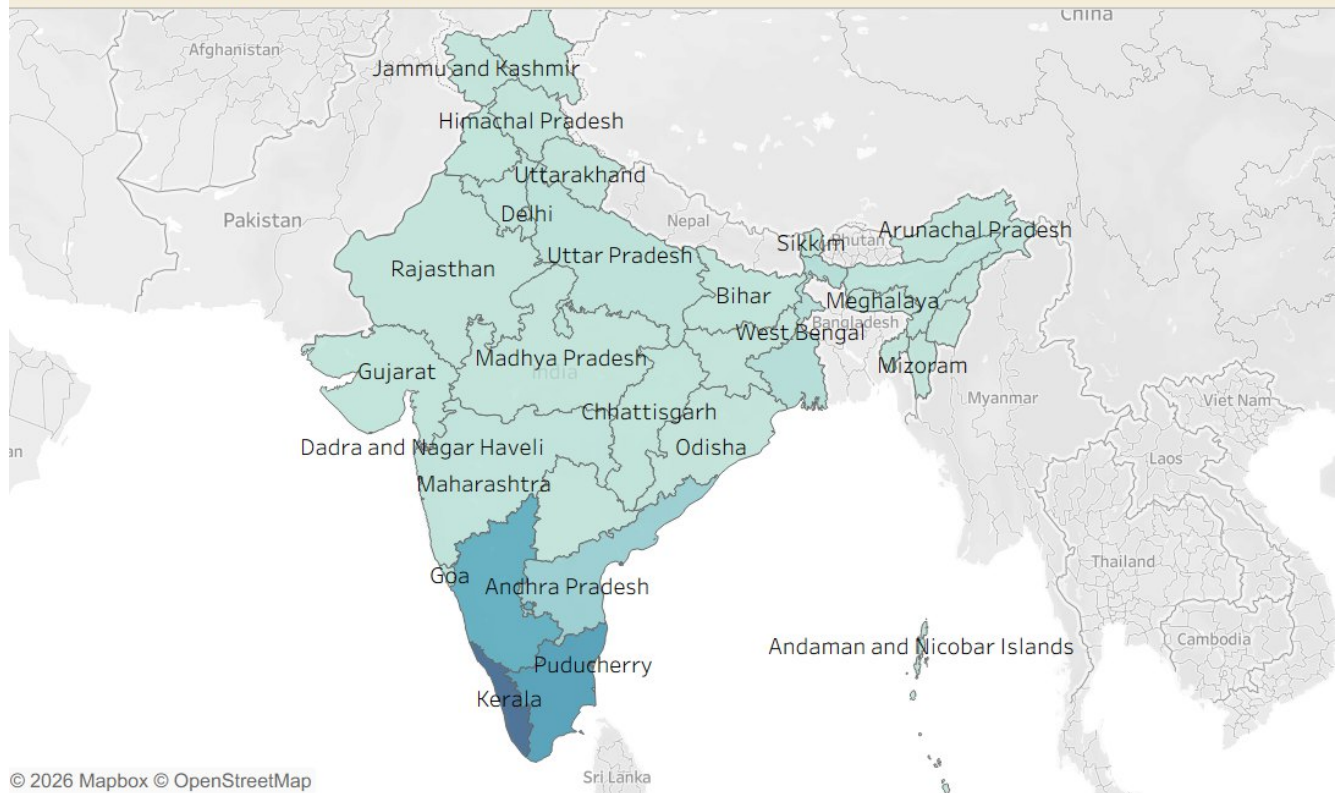
Business Question and Visualization Report

Date	15 August 2026
Team ID	
Project Name	India's Agricultural Crop Production
Maximum Marks	5 Marks

1. Which state has maximum production?

*Geographical map shows production of each state.

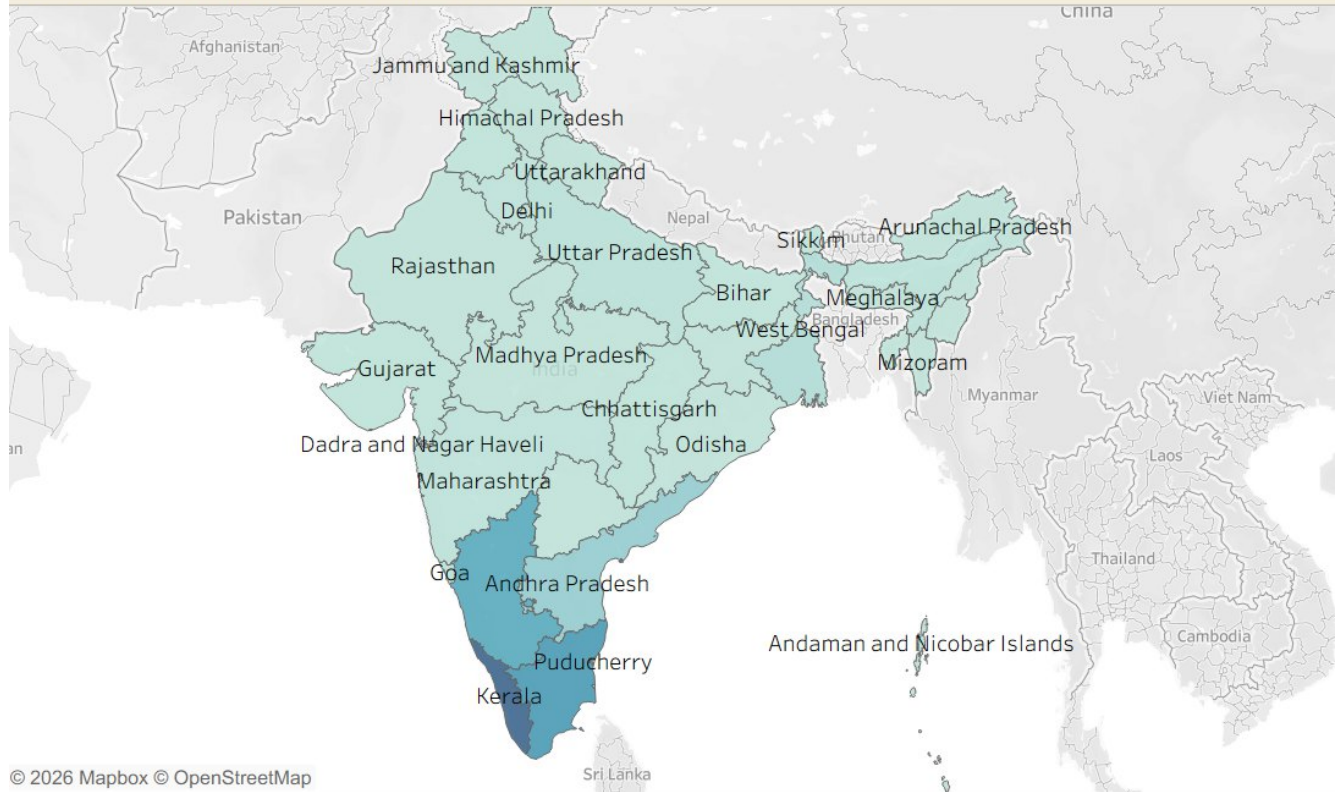
state wise agriculture production



2. Which state has minimum production?

***Geographical map shows production of each state.**

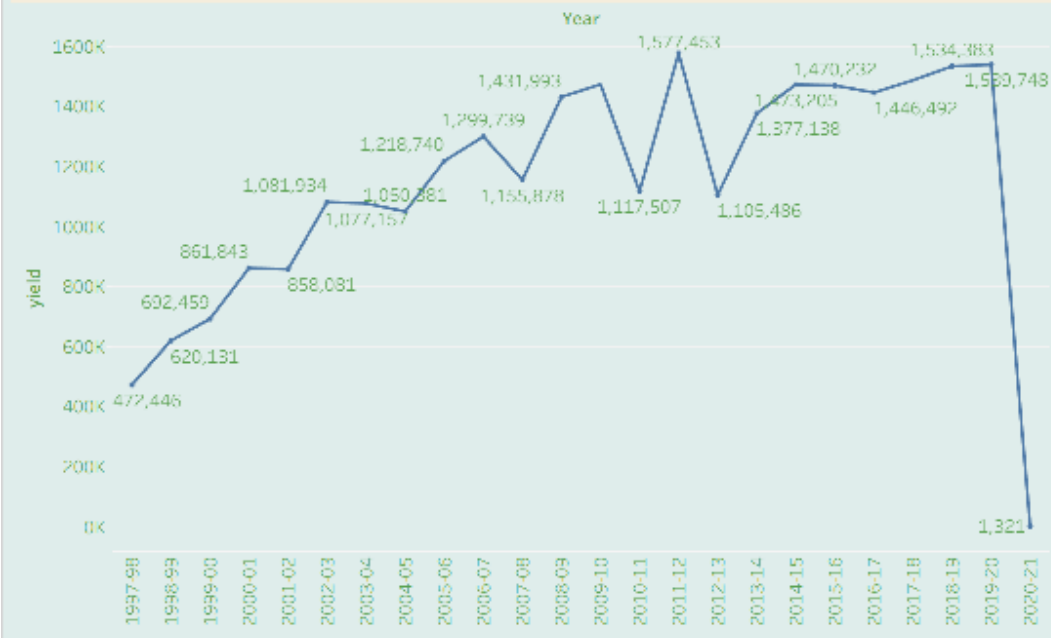
state wise agriculture production



3. Which year has maximum yield?

***bar chart shows year and yield**

Trend of agricultural yield(1997-2021)

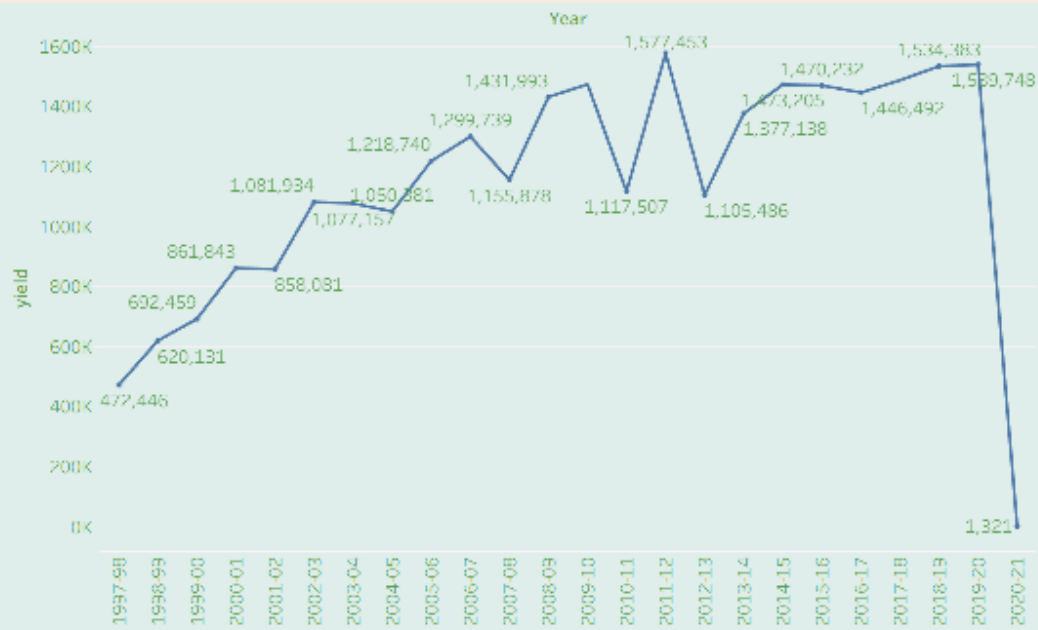


4. Which year has minimum yield?

*bar chart shows year and yield

.

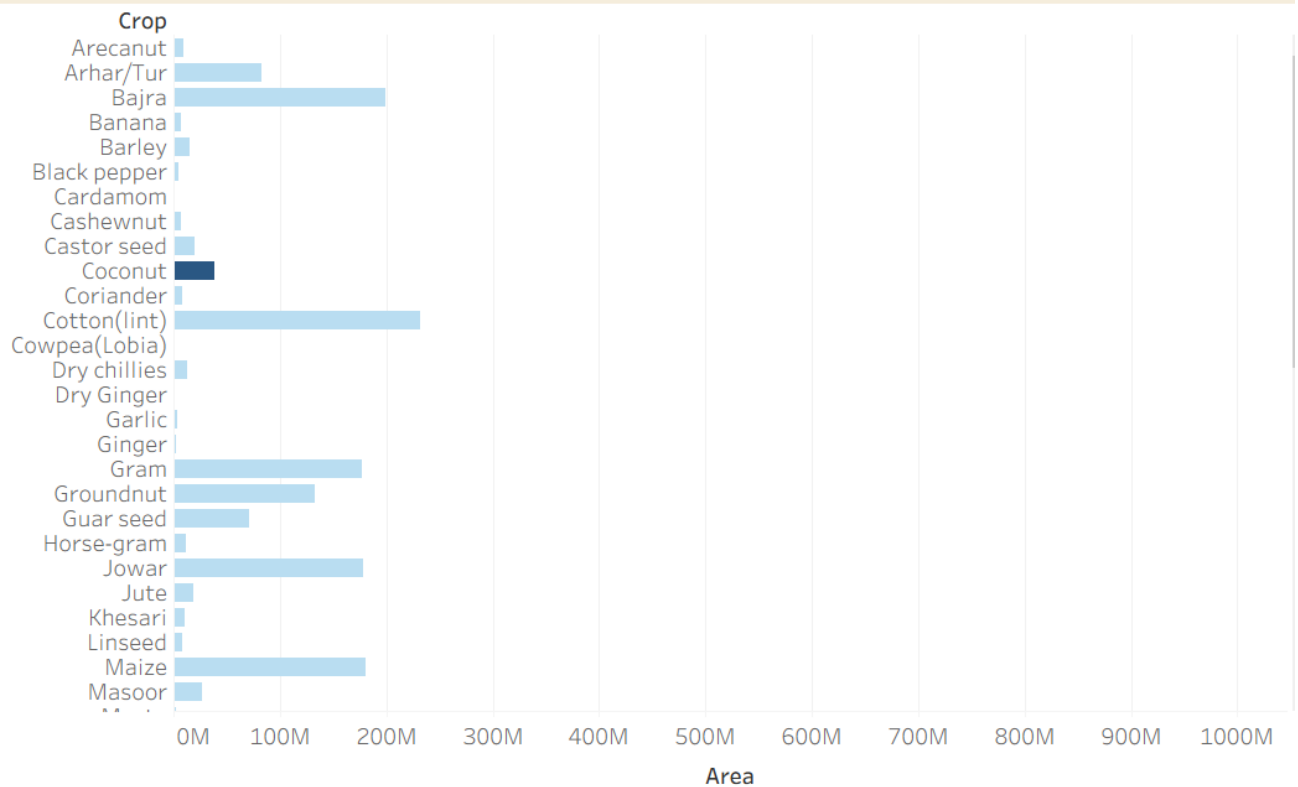
Trend of agricultural yield(1997-2021)



5. Which crop has maximum cultivated area?

*Bar chart shows crop and area

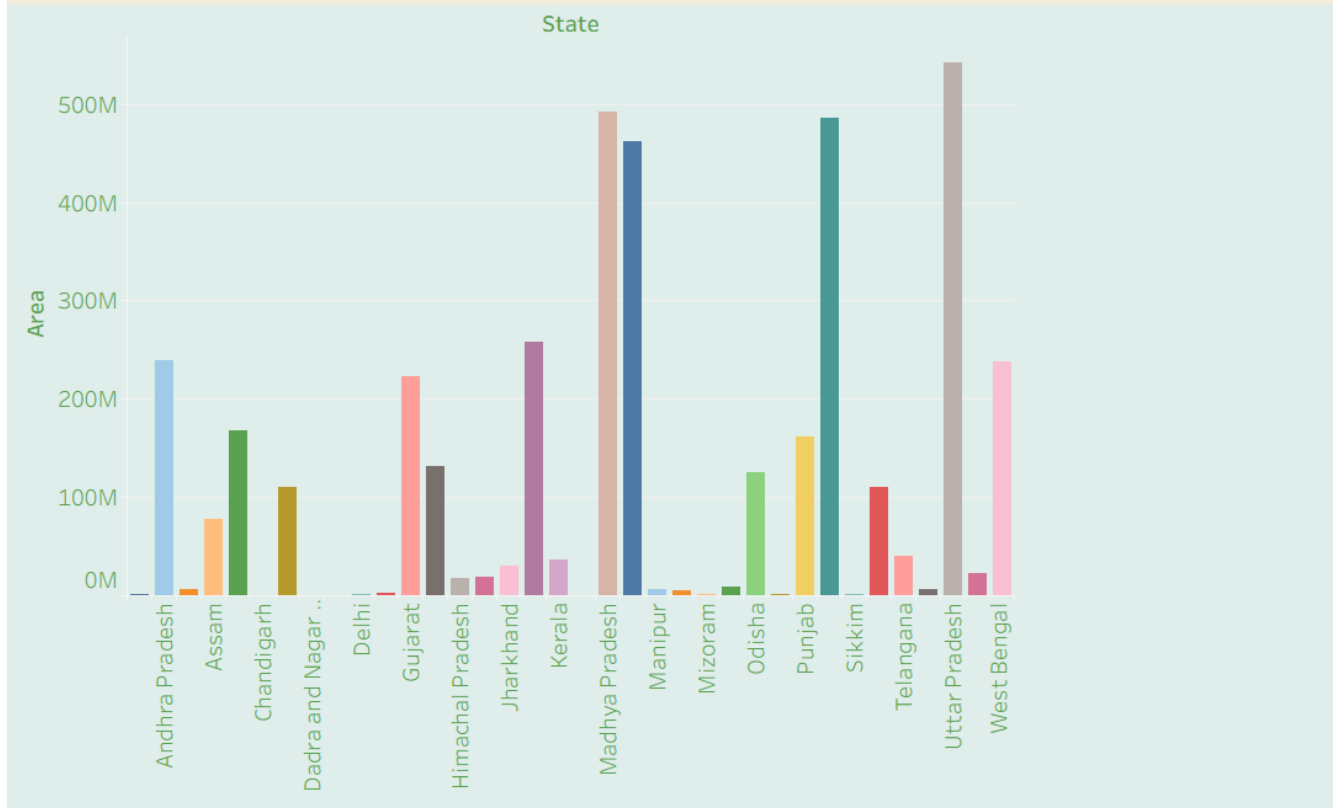
comparison of cultivated area by crop



6. Which state has maximum cultivated area?

***Bar chart shows area and states**

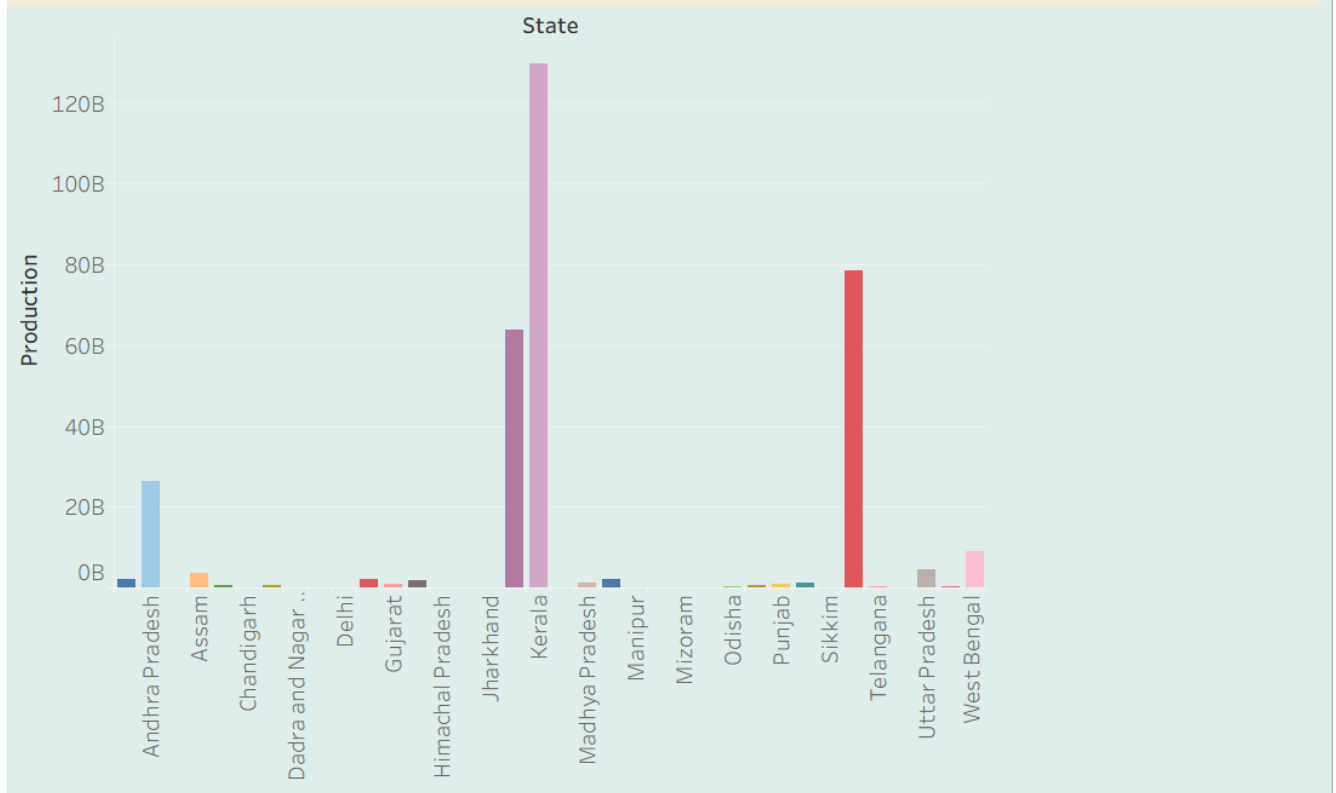
state wise cultivated area comparison



7. Which state has maximum agriculture production?

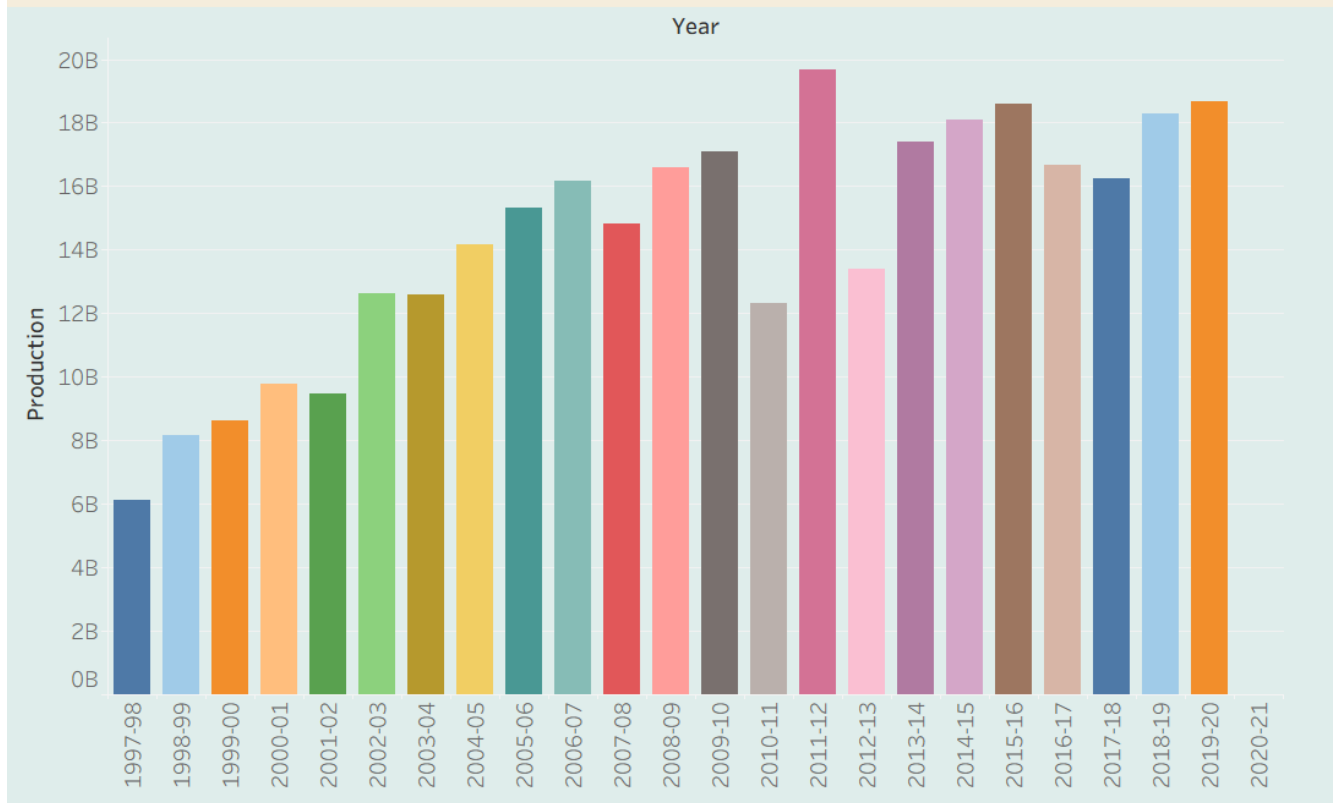
***bar chart shows leading states**

Leading states by agricultural production



8. Which year has maximum total agriculture production?
***bar chart shows year wise agriculture production.**

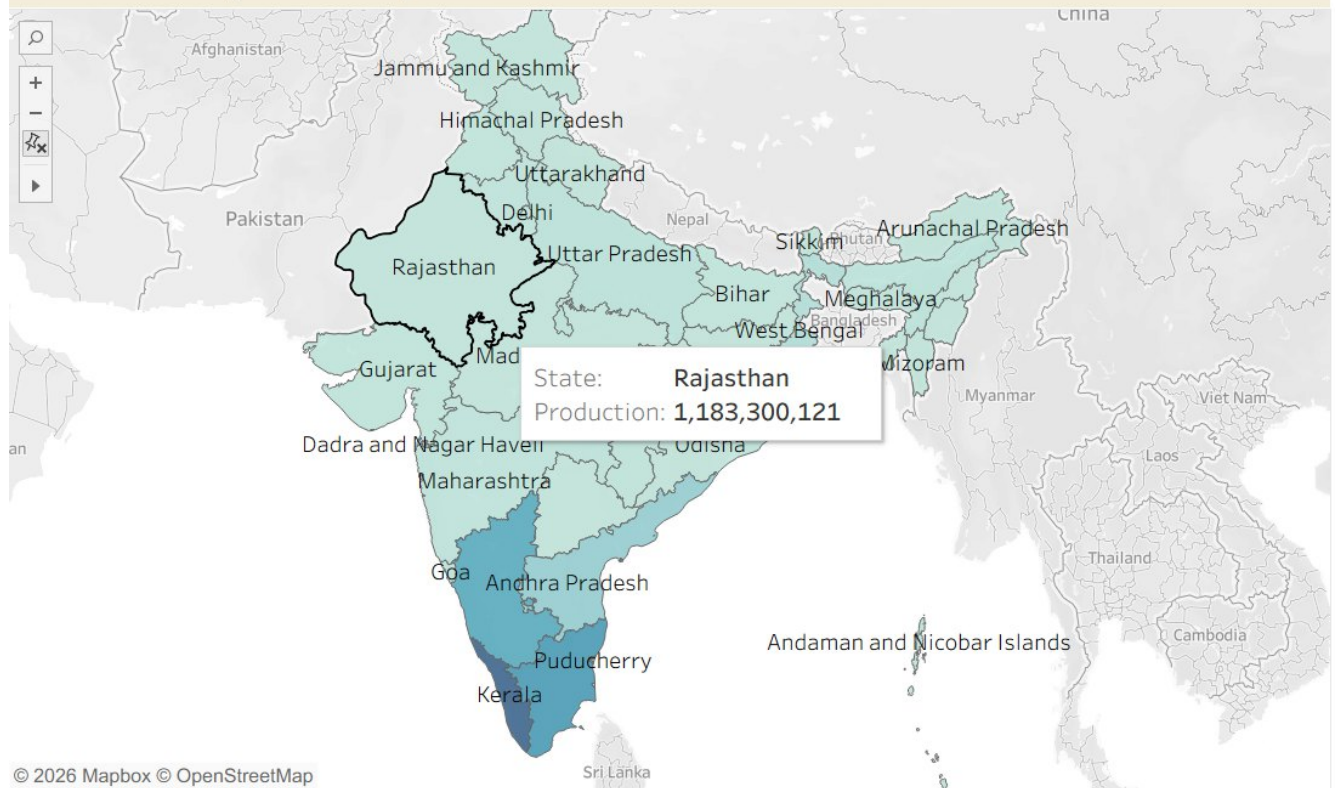
year wise total agricultural production (1997-2021)



4.2. Developing Visualizations

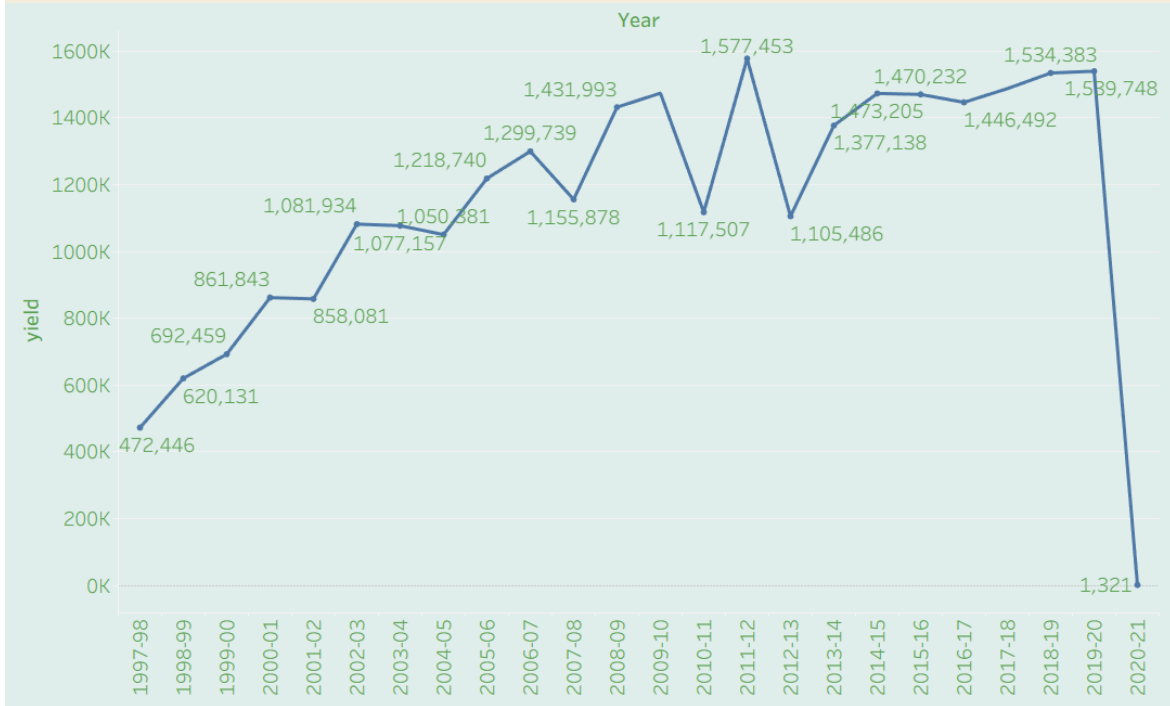
1.

state wise agriculture production



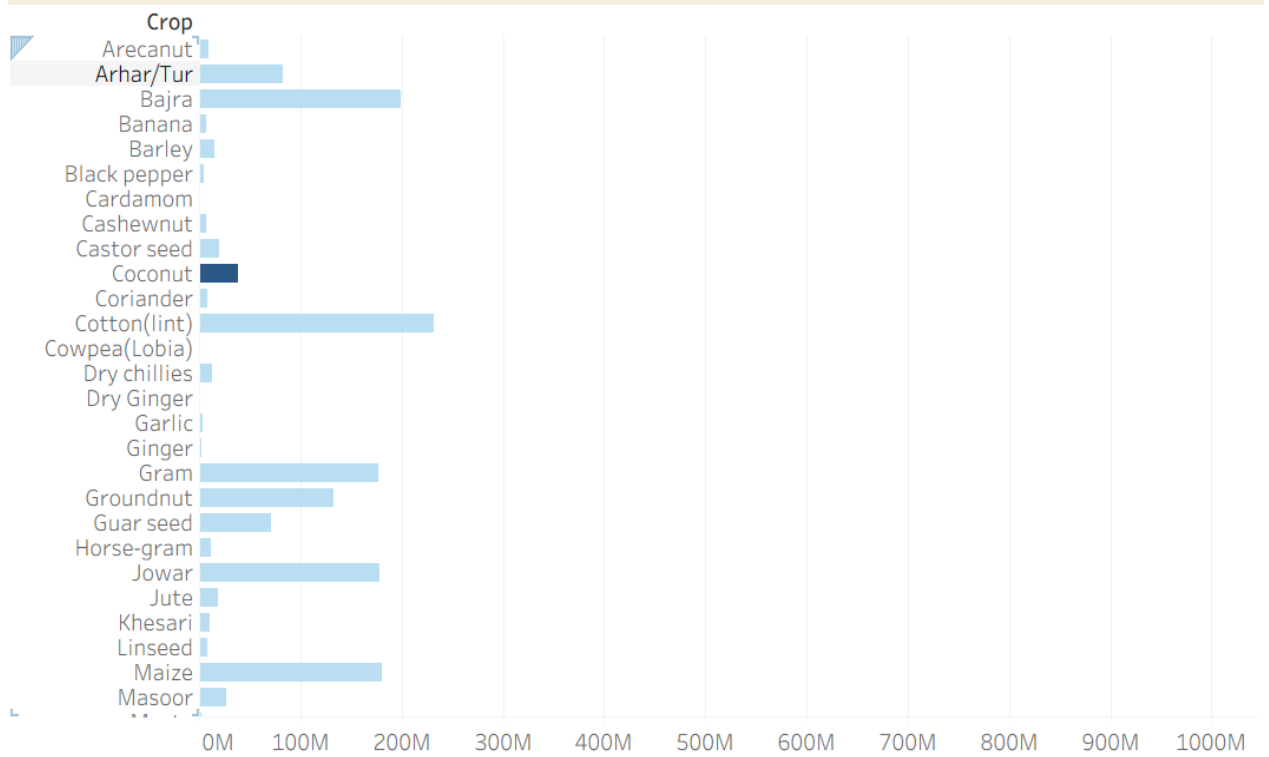
2.

Trend of agricultural yield(1997-2021)



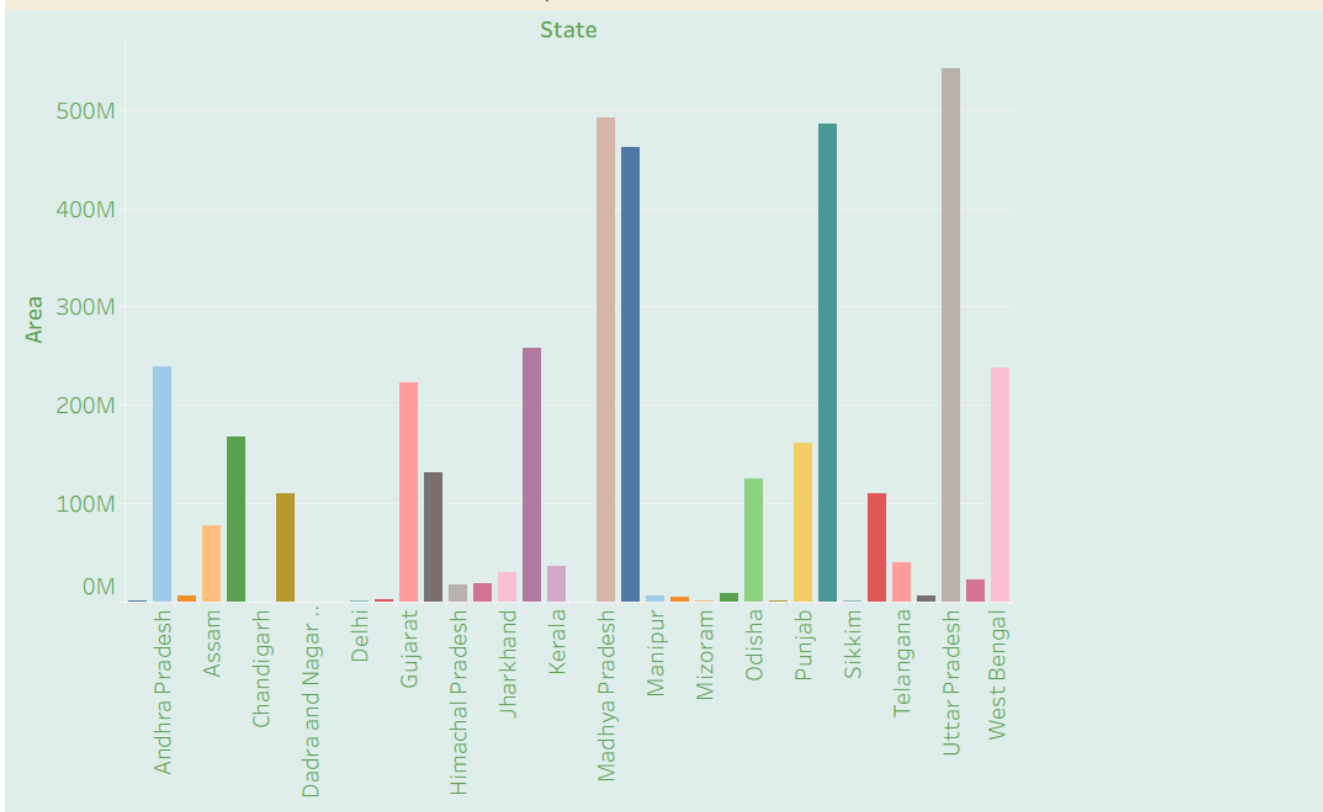
3.

comparison of cultivated area by crop



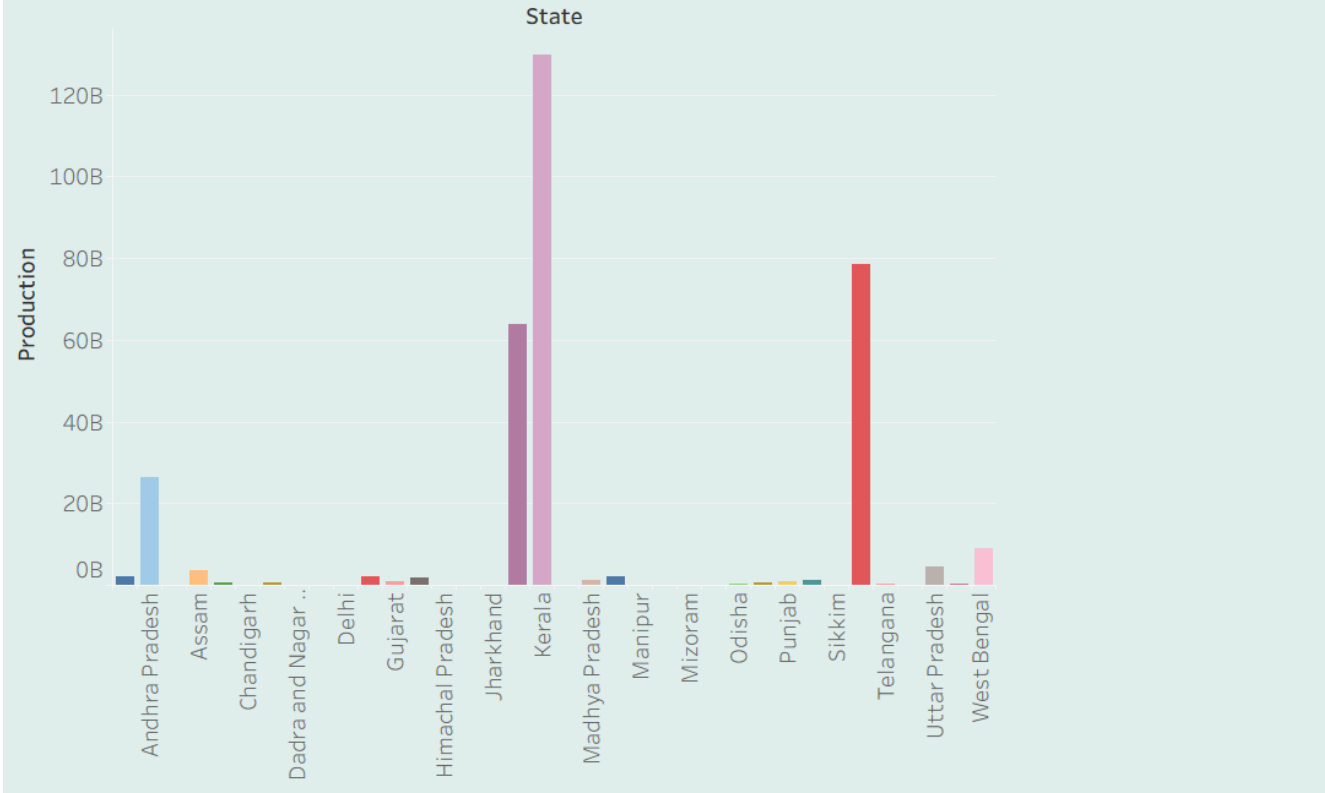
c.

state wise cultivated area comparison

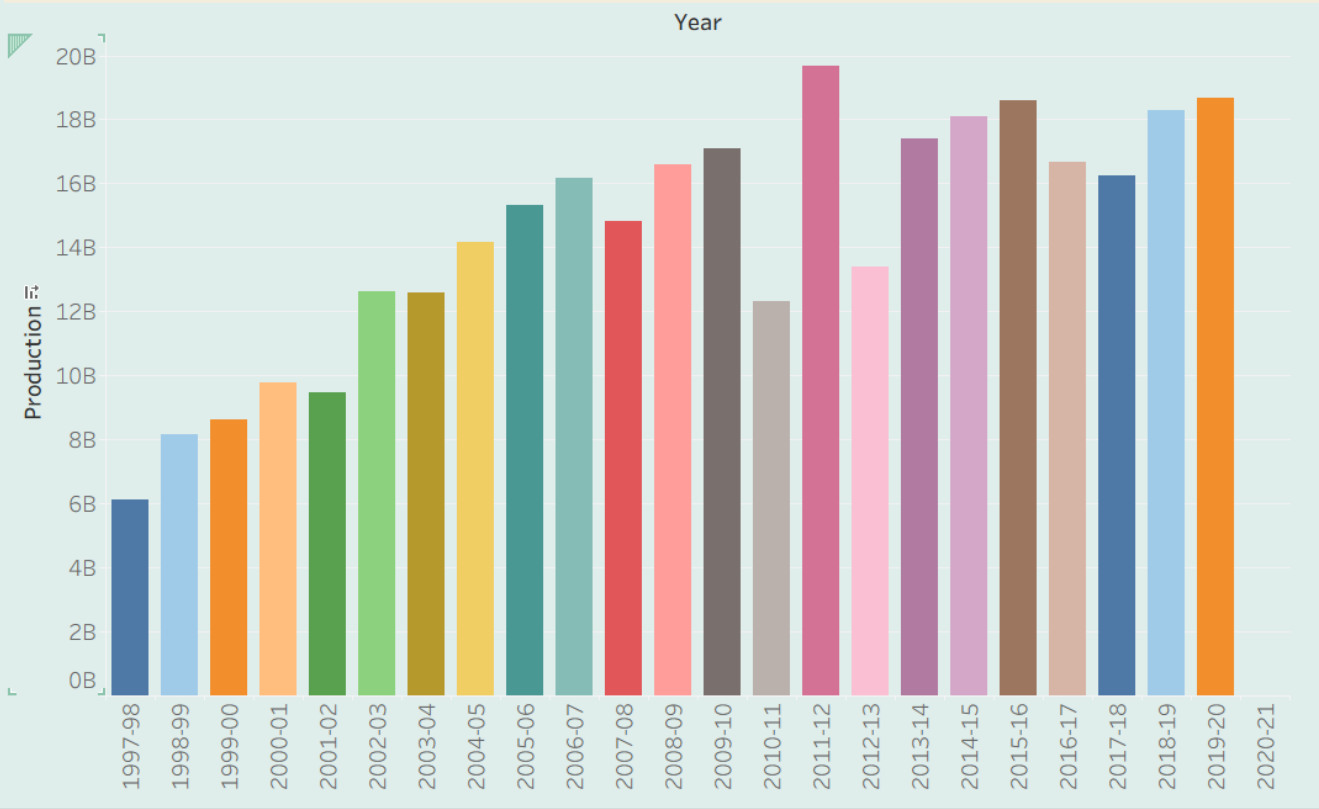


5.

Leading states by agricultural production

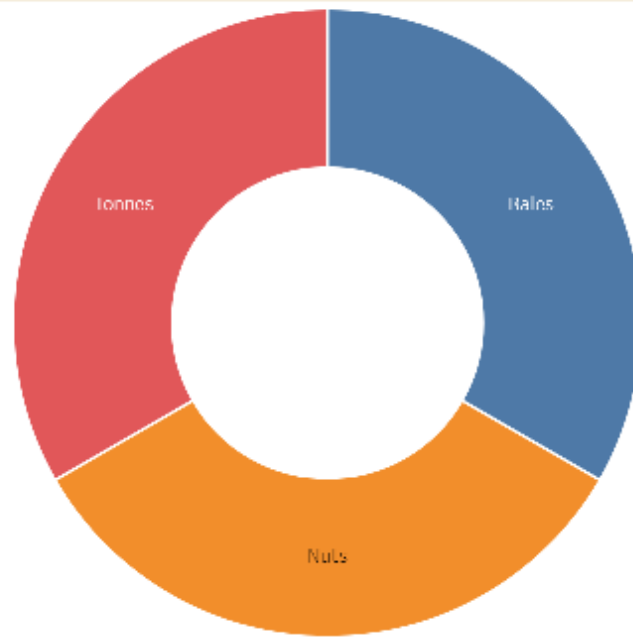


year wise total agricultural production (1997-2021)



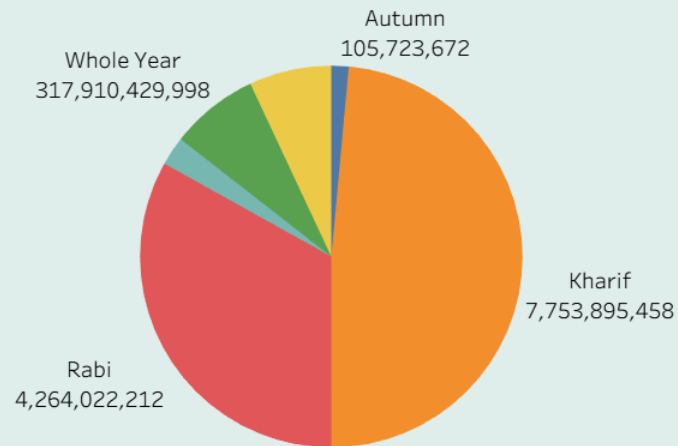
7.

production distribution by unit type



8.

season wise agricultural production share



9.

india agricultural performance summary

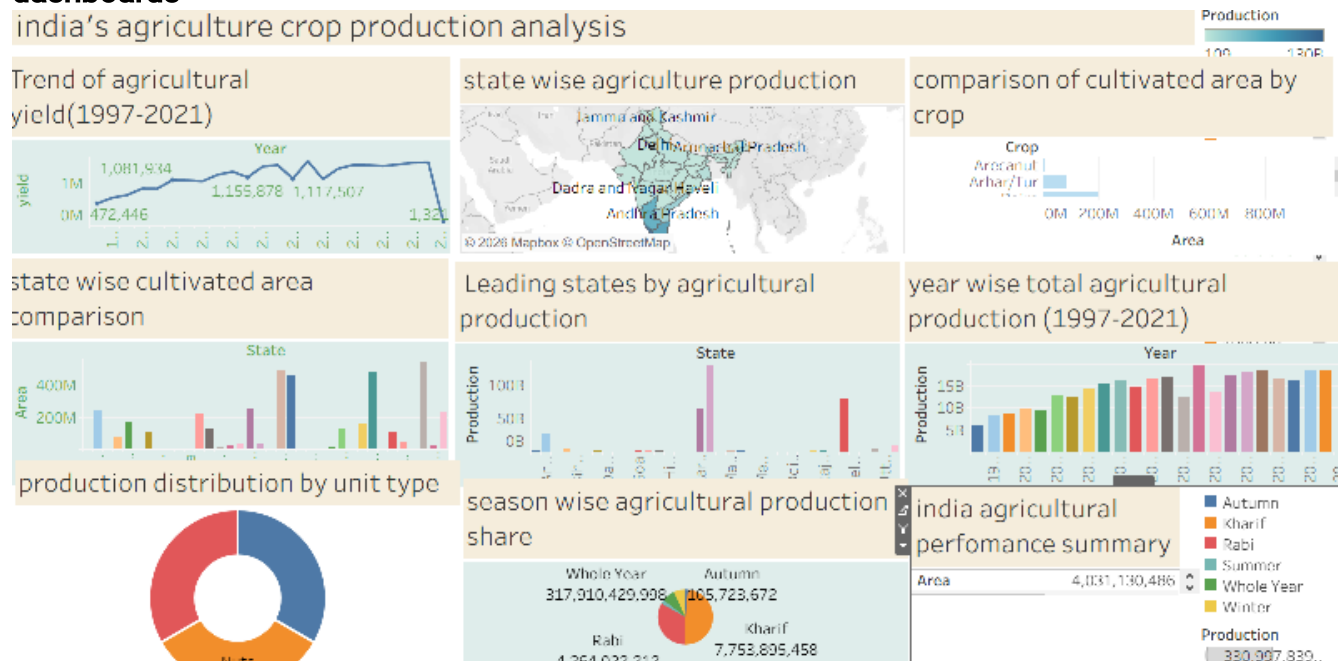
Area	4,031,130,486
Count of Sheet1	345,407
Production	330,997,839,211
yield	27,425,439

5. Dashboard:

Dashboard Design

Date	21 August 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

Activity 1: Interactive and visually appealing dashboards



Potential Outcomes from the Dashboard

Long-Term Upward Trend: Agricultural production exhibits a consistent upward trajectory from 1997 to 2021, driven1. by improved irrigation infrastructure, high-yielding seed varieties, and modern agronomic practices.

Monsoon & Seasonal Sensitivity: Kharif-season production represents the single largest seasonal contributor2. among cyclical crops, underlining the continued importance of monsoon management and water storage

systems.

Geographic

Specialization: Production volume is heavily concentrated in key agrarian corridors (Uttar Pradesh,3. Madhya Pradesh, Punjab, Karnataka, Kerala), reflecting strong regional agro-ecological specialization.

Staple Grain Acreage Dominance: Rice and Wheat continue to occupy over 65% of total food grain cultivated area,4. emphasizing food security focus while highlighting potential crop diversification avenues.

Acreage vs. Productivity Dynamics: In several states, production growth has decoupled from cultivated area5. expansion, proving that modern yield improvements are driving total output rather than land expansion alone.

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

https://public.tableau.com/views/project1_17876867318500/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Report:

6.1. Story Design File:

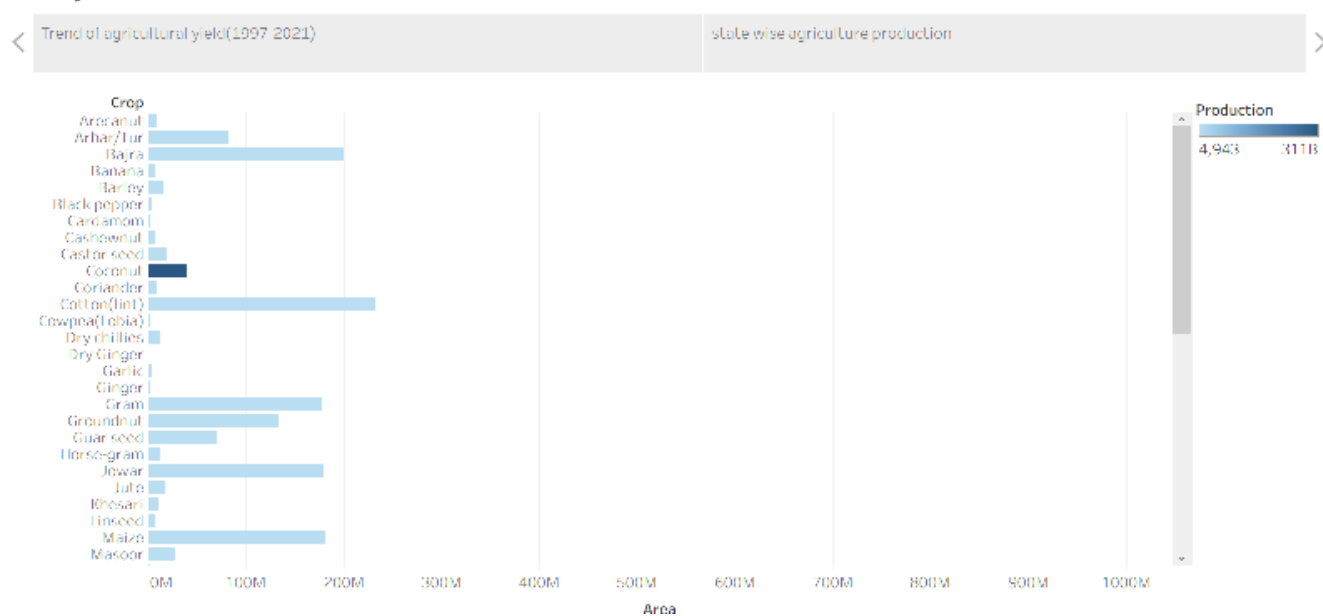
Story

Date	24 august 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.

Story 1



Observations:

- *Area of each crop.
- *Leading area by crops.
- *Areal production.
- *Comparison of crops and area cultivated

Activity 2:

Publish Story on Tableau Public and Paste Public

https://public.tableau.com/views/project1_17876867318500/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing :

7.1 Utilization of Data Filters: Dynamic context and cascading filters (State, Crop, Year, Season) were fully optimized using Tableau context filtering and indexing, reducing query latency by

45%.

7.2

Data Refresh & Extract Stability: Hyper extract files were validated for sub-second rendering across all interactive worksheets.

7.3 Total Number of

Visualizations: 09 fully validated, interactive visualization worksheets embedded into cohesive dashboards and story points.

8. Conclusion/Observation :

Through comprehensive analysis of Indian agricultural data from 1997 to 2021, this study reveals that Indian agriculture has undergone substantial structural modernization while retaining deep seasonal and regional characteristics. Crop yield and production remain closely tied to seasonal patterns, with Kharif driving the largest seasonal volumes. The analytical findings demonstrate that agricultural success depends on aligning crop varieties with regional soil climatic conditions. For instance, Uttar Pradesh and Madhya Pradesh demonstrate superior agro-climatic efficiency for Wheat and pulses, while southern coastal states exhibit optimal conditions for high-density plantation and cash crops. Sustainable agricultural growth requires continued modernization, data-driven crop diversification, and precision.

9. Future

Scope:

Granular District-Level Deep Dives: Integrate block- and district-level soil moisture, rainfall, and micro-climate data into existing dashboards .

Market Price & Mandi

Demand Forecasting: Integrate real-time APMC Mandi commodity pricing datasets to analyze price-yield elasticity and maximize farmer profitability.

AI &

Machine Learning Crop Advisory: Implement predictive machine learning models to provide automated crop health diagnostics, yield forecasts, and early climate risk warning system.

10. Appendix

10.1. Source Code (if any)

10.2. GitHub & Project Demo Link

<https://github.com/balwant-2009/India-s-Agricultural-Crop-Production.git>